SynchroKin Clinical Dashboard

Official User & Operations Guide

Contents

1 Patient Setup: Strapping the Hardware

Before launching the software, the physical hardware nodes must be strapped to the patient. Proper orientation of the sensor casings is critical for accurate 3D mapping.

Orientation Rules:

- **Alpha (Chest) Node:** Strap securely to the center of the chest. The DPDT power switch must be facing towards the **left side** of the patient's body.
- **Limb Nodes (Bicep/Forearm):** Strap securely to the outer side of the respective limbs. The DPDT power switch must be facing **upwards** (towards the patient's head) when the arms are resting at their sides.

Figure 1: Correct hardware mounting orientations. Note the direction of the DPDT switches.

2 Launching & Connecting the System

The SynchroKin system receives wireless data through a Master ESP32 plugged into the computer via USB. You must identify the correct COM port before tracking can begin.

2.1 Identifying the COM Port

1. Plug the Master ESP32 Receiver into a USB port on your Windows PC.
2. Open the Windows Start Menu, type **Device Manager**, and hit Enter.
3. Expand the dropdown labeled **Ports (COM & LPT)**.
4. Plug your Master ESP32 and note down which new port appeared (e.g., `COM3`, `COM13`). Enter this in the dashboard of the app.

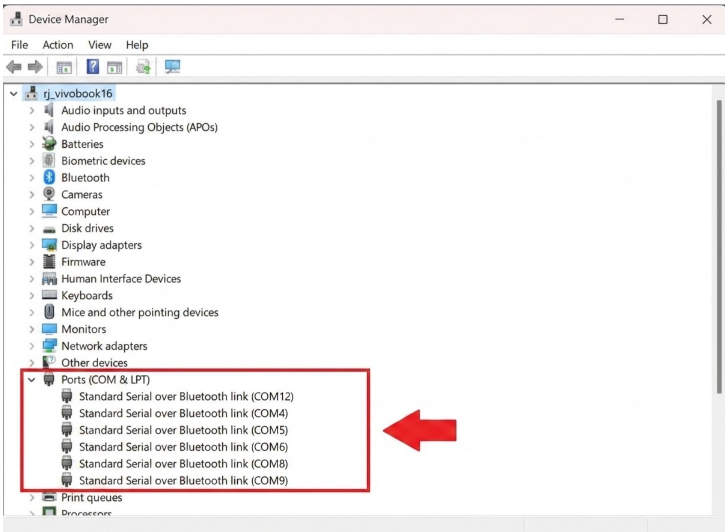

Figure 2: Locating the assigned COM port inside Windows Device Manager.

2.2 Connecting the Dashboard

1. Launch the **SynchroKin Dashboard** executable (`.exe`).
2. Once the main dashboard screen loads, locate the **COM Port Input Field**.
3. Enter the COM port you found in Device Manager (e.g., type `COM13`) and press Enter. The system will automatically open the data pipeline.

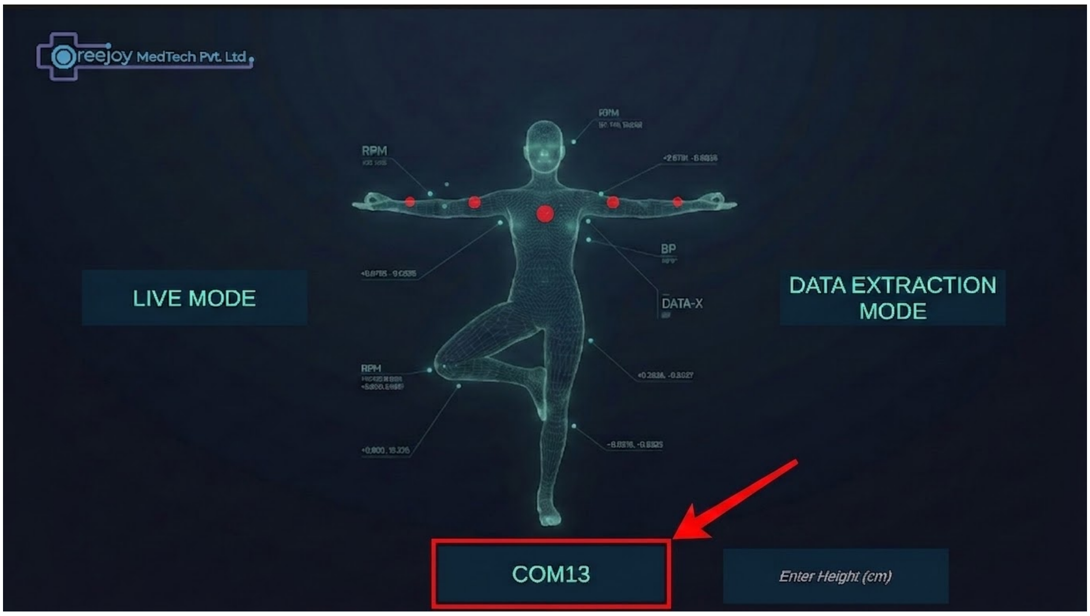

Figure 3: Entering the COM port on the main SynchroKin dashboard.

3 Dynamic Node Routing

Because any physical module can be strapped to any limb, you must tell the software which module goes where.

1. Power on all the patient's wearable nodes using their DPDT switches.
2. On the App Dashboard, look at the colour of the nodes. Red means either the node has "None" selected or the module assigned to that node has not been turned on. They will turn blue as modules successfully connect.
3. Use the dropdown menus on each digital limb to select the corresponding physical node (e.g., Assign "Beta" to the Right Bicep, "Gamma" to the Left Bicep).
4. Any limb assigned to "None" will remain locked in a natural resting pose.

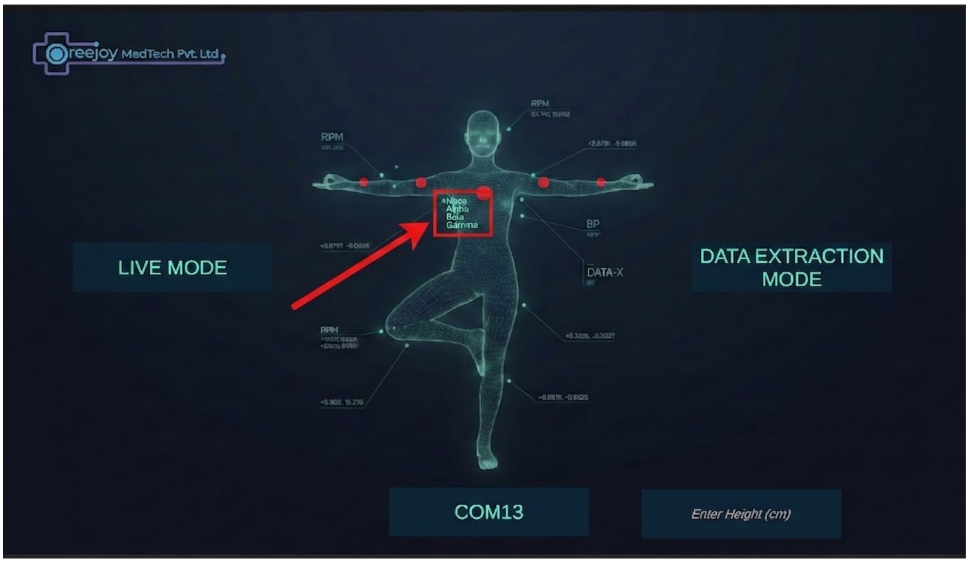

Figure 4: Mapping physical Greek nodes to their respective digital limbs using the UI dropdowns.

4 Live Mode: Validation & The "Snap" Calibration

Before recording a clinical test, it is highly recommended to enter **Live Free Play Mode** to ensure the IMUs have booted up with the correct spatial hemisphere mapping.

1. Click **Live Mode** on the dashboard.
2. Ask the patient to slowly lift their right arm. The 3D avatar should perfectly mimic this movement.
3. **Troubleshooting a Flipped Axis:** If the digital arm bends backwards or sideways while the patient's arm is straight, the IMU booted in the wrong quadrant.
4. **The "Snap" Fix:** Do not restart the system. Simply detach that specific module from the patient's arm. Hold it in your hand and continuously rotate it around on all axes. You will suddenly see the 3D avatar's arm "snap" into the correct, forward-facing alignment.
5. Once it snaps, keep it in that orientation and immediately strap it back onto the patient. Repeat this for any other limbs behaving erratically.

5 Data Extraction Mode (Clinical Testing)

To record timestamped kinematic data (such as Postural Sway metrics), return to the main dashboard and enter **Data Extraction Mode**.

5.1 Calibration Phase (The T-Pose)

1. Enter the patient's height (in cm) into the scaling input field to size the avatar correctly.
2. Select your desired clinical test (e.g., Postural Sway Assessment).
3. The screen will display **Auto-Calibration**. Instruct the patient to stand completely still in a rigid **T-Pose** (arms raised parallel to the floor).
4. They must hold this T-Pose for the **3-second countdown**. The system uses this snapshot to mathematically zero-out the suit offsets.
5. Once the Action Phase begins, the patient may perform the required test movements.

6 Exporting Clinical Reports

During the Action Phase, the software actively records 100Hz kinematics and calculates live Root Mean Square (RMS) sway values.

1. When the physical assessment is complete, click the **End Recording** button.
2. You will be brought to the Clinical Report screen, displaying the final Anterior-Posterior (Pitch) and Medial-Lateral (Roll) peak and RMS metrics.
3. Use the buttons at the bottom of the screen to save the data:
   - **Export CSV:** Saves the frame-by-frame raw quaternions and Euler angles to a spreadsheet.
   - **Export JSON:** Saves the same raw data into a structured format for database integration.
   - **Export Graph:** Renders and downloads a high-resolution PNG image of the live sway chart generated during the test.